

Position Generalization in Translated FashionMNIST

Architecture, patch scale, and patch embedding

Final English report

Controlled experiments · seed 42

Abstract. We test whether image classifiers retain accuracy when object position changes between training and testing. Under one shared training recipe, CNN records the highest accuracy and reaches 35.40% on the hardest fixed-to-random transfer. Patch size 8 leads three of four ViT settings. Equivalent Conv2d and Flatten + Linear projections remain within 0.61 percentage points in this run.

1 Question

A classifier may learn class appearance and object location together. This study isolates the effect of location by comparing random-position images (A) with centered images (B).

Setting	Training	Testing	Purpose
A -> A	random	random	in-distribution reference
B -> B	center	center	fixed-position reference
A -> B	random	center	random-to-fixed transfer
B -> A	center	random	hardest position shift

2 Controlled protocol

Each configuration trains one model on A and one on B. Validation chooses the checkpoint; the official test set is evaluated only after selection.

Item	Controlled choice
Data	FashionMNIST on a 64 x 64 black canvas
Split	90% train, 10% validation; official test set held out
Training	15 epochs, AdamW, seed 42
Selection	best validation checkpoint
Reporting	test accuracy and elapsed training time

Primary measure. B -> A is the hardest setting because a model trained only on centered objects must classify objects across the canvas.

Architecture and Position Shift

MLP, CNN, and patch-8 ViT under one evaluation protocol

3 Accuracy

Configuration	Parameters	A -> A	B -> B	A -> B	B -> A
MLP	542,474	71.89	89.57	71.97	13.99
CNN	205,994	92.35	93.14	92.87	35.40
ViT, patch 8	811,402	80.40	88.27	81.13	18.81

Table 1. Test accuracy (%). The best value in every column belongs to CNN.

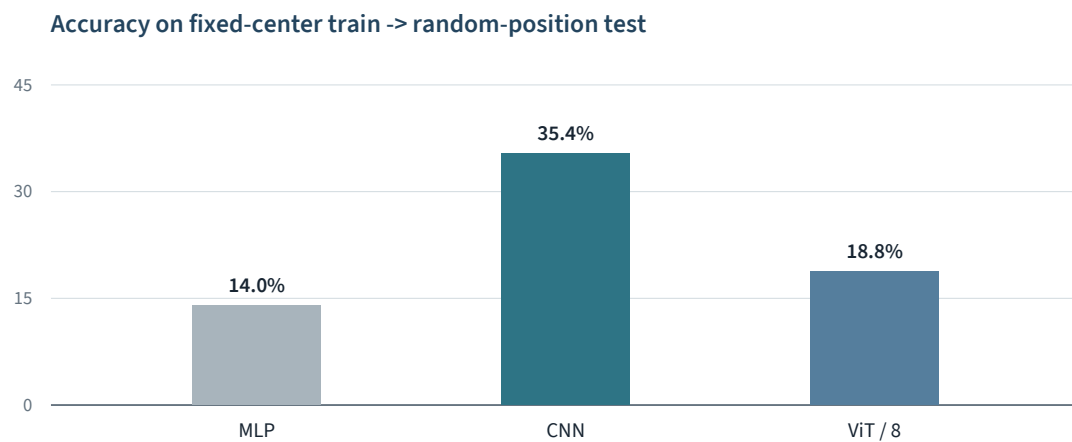


Figure 1. CNN retains substantially more accuracy on B -> A.

BEST B -> A 35.40% CNN	CNN ADVANTAGE +16.59 pp over patch-8 ViT	SMALLEST MODEL 205,994 CNN parameters
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4 Effect of the position shift

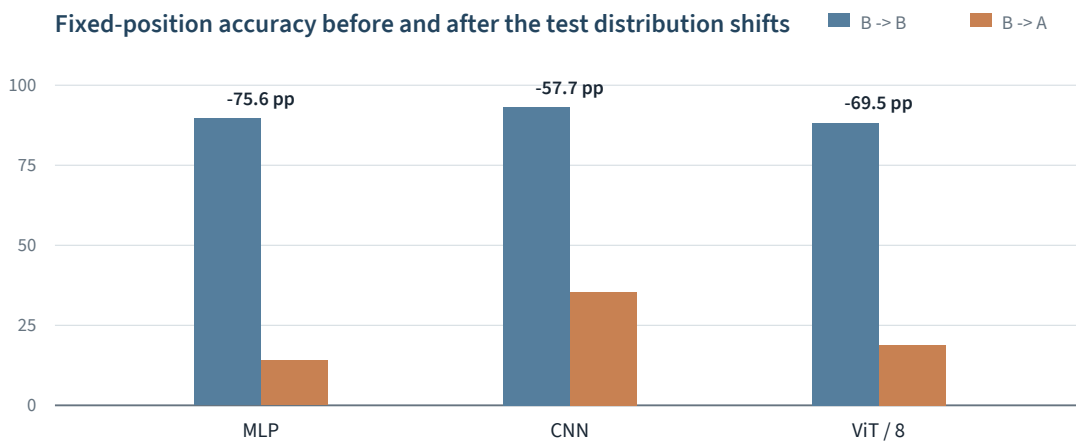


Figure 2. All models degrade after centered training is tested at random positions.

Patch Scale and Embedding

Two controlled changes to the shared Transformer configuration

5 Patch scale

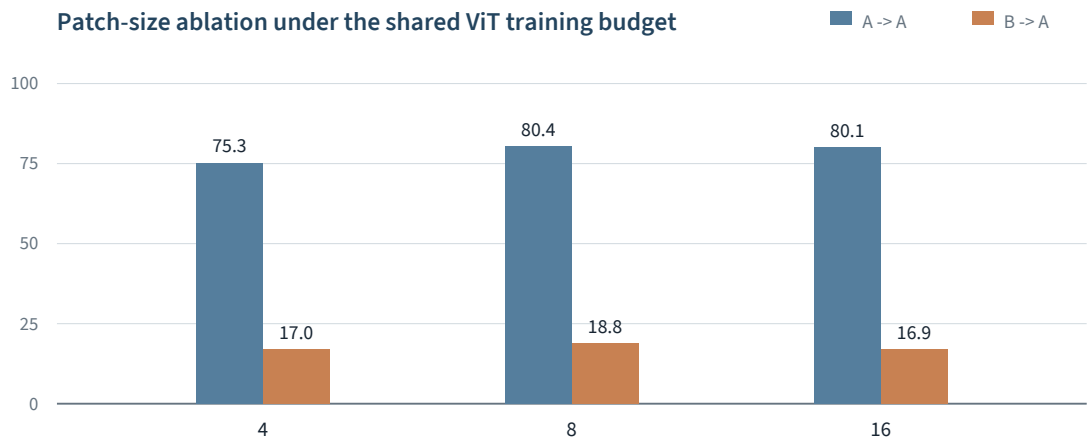


Figure 3. Patch size 8 leads three of four evaluated settings.

Patch	Tokens	A -> A	B -> B	A -> B	B -> A	Time
4	256	75.28	87.22	74.97	17.02	8.93 min
8	64	80.40	88.27	81.13	18.81	7.36 min
16	16	80.07	88.87	80.97	16.91	6.46 min

Table 2. Accuracy (%) and total time for the A- and B-trained models.

Result. Patch size 8 leads on A -> A, A -> B, and B -> A. Patch size 4 raises the token count to 256 and is slower without an accuracy gain.

6 Patch embedding

For non-overlapping patches, a strided Conv2d and a shared linear layer perform the same type of patch-wise projection. Their results remain close.

Test	Conv2d	Flatten + Linear	Absolute gap
A -> A	80.07	79.67	0.40
B -> B	88.87	88.80	0.07
A -> B	80.97	80.48	0.49
B -> A	16.91	16.30	0.61

Table 3. Test accuracy (%) for patch size 16. Maximum absolute gap: 0.61 points.

Conclusions and Reproducibility

Concise interpretation and a documented reproduction path

7 Conclusions

- 01** CNN performs best under the shared recipe. It leads all four settings and retains the most fixed-to-random accuracy.
- 02** Patch size 8 leads three of four settings. It gives the best B -> A result without the 256-token cost of patch size 4.
- 03** Equivalent embeddings stay close in this run. Conv2d and Flatten + Linear differ by at most 0.61 percentage points.

8 Limitations

Scope

- One training and placement seed; no variance estimate.
- Model sizes and optimal hyperparameters are not matched.
- Runtime is hardware- and order-specific.

Interpretation rule

Small differences are described only as numerically close in this run; no significance claim is made.
Conclusions apply to the shared protocol rather than all possible training recipes.

9 Reproducibility

Check	Recorded value
Code	translated_fashionmnist/experiments
Metrics	results/metrics.csv
Seed / epochs	42 / 15
Dependency lock	requirements-lock.txt
Environment	Python 3.12.13, PyTorch 2.11.0, CUDA 12.8
Hardware	NVIDIA GeForce RTX 5070 Laptop GPU

```
python -m translated_fashionmnist.experiments.compare --groups all --download
```

10 Implementation map

Component	Location	Role
Dataset	data.py	construct A and B canvases
Models	models.py	MLP, CNN, ViT, patch embeddings
Protocol	experiments/protocol.py	train, validate, and evaluate
Engine	engine.py	shared training and evaluation primitives
Runner	experiments/compare.py	execute all three comparison groups
Outputs	results/	metrics, history, and figures